

Kalman filter - for linear systems

Sensors and Systems

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Outline

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- ▶ *State estimation* is part of the ES8 course: Sensors and Systems.
- ▶ State estimation is used for estimating something which is not directly measured, or which is measured with lots of noise, from something else which is measured.
- ▶ State estimates can be used for diagnostics, fault detection or control as e.g. state space control.
- ▶ State estimators are based on dynamic models relating the state and measurement.
- ▶ State estimators can be based on *deterministic models* and are then called observers.
- ▶ Here the state estimators are based on *stochastic models*.

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Literature and lectures

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- ▶ Good literature for this course is hard to find.
- ▶ We will use: Mohinder S. Grewal and Angus P. Andrews, Kalman Filtering, Wiley, 2008 and also some lecture notes and papers.
- ▶ The lectures will not follow the book closely but will try to give alternative expositions as well and sometimes use other notation.
- ▶ The focus is on discrete time models. Continuous time models will only be briefly mentioned.

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State estimation problem for linear systems

General formulation

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Terminology

Given a state space model and *input/output* measurement up to and including time t estimate the *state* at time s .

The estimation are given specific names depending on the relation between t and s .

$s < t$ State *smoothing*.

$s = t$ State *filtering*.

$s > t$ State *prediction*.

The discrete-discrete (DD) estimation problem

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Assume the system:

$$x_{k+1} = \Phi_k x_k + \Gamma_k u_k + w_k, \quad w_k \in \text{NID}(\underline{0}, Q_k),$$

$$y_k = H_k x_k + D_k u_k + v_k, \quad v_k \in \text{NID}(\underline{0}, R_k),$$

$$E(w_k v_l^T) = \underline{0}$$

Given measurements and initial values:

$$\mathcal{Y}_k \triangleq y_0, y_1, y_2 \dots y_k, \quad u_0, u_1, u_2 \dots u_k$$

$$x_0 \in N(\hat{x}_0, P_0)$$

Find: $\hat{x}_k$ which minimized the mean square error

$$E((x_k - \hat{x}_k)^T M (x_k - \hat{x}_k)), \quad M \geq 0$$

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The state model is in continuous time i.e. a stochastic differential equation (SDE):

$$dx(t) = (F(t)x(t) + B(t)u(t))dt + d\omega(t) ,$$

$$\omega(t) \in W(Q(t)) ,$$

$$y(t_k) = H_k x(t_k) + D_k u(t_k) + v_k , v_k \in \text{NID}(\underline{0}, R_k)$$

Given measurements and initial values:

$$\mathcal{Y}_k \triangleq y(t_0), y(t_1), y(t_2) \dots y(t_k) , u(s), t_0 \leq s \leq t_k$$

$$x(t_0) \in N(\hat{x}(t_0), P(t_0))$$

Find: $\hat{x}_k$ which minimized the mean square error

$$E((x_k - \hat{x}_k)^T M (x_k - \hat{x}_k)) , M \geq 0$$

Using standard calculus for ODE the SDE can be rewritten as follows:

$$dx(t) = (F(t)x(t) + B(t)u(t))dt + d\omega(t) \Rightarrow$$

$$\dot{x}(t) = F(t)x(t) + B(t)u(t) + w(t) ,$$

$$w(t) = \frac{d\omega(t)}{dt}$$

- ▶ This simplified formulation is wrong from a strict mathematical point of view.
- ▶ To make any sense $w(t)$ must be *continuous white noise* which is a unphysical signal with constant spectrum for all frequencies and consequently an infinite variance.

The continuous-continuous (CC) estimation problem

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- ▶ The problem where continuous measurements is assumed is called the continuous-continuous estimation problem and the solution is called the Kalman-Busy filter.
- ▶ In practice the measurements are always in discrete time.
- ▶ Continuous measurements with continuous time white noise is also unphysical in some sense.

Notice: You do not need to worry about continuous time white noise and details on SDE's in this course.

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Estimators based on probabilistic models

Simplest possible problem

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Estimation from a one dimensional distribution

Given the density function $f(x)$ for a scalar stochastic variable x find the estimate $\hat{x}$ that minimize the mean square error (MSE) $E((x - x_e)^2)$ where x_e is any estimate.

$$\begin{aligned} L(x_e) &= E((x - x_e)^2) \\ &= E(((x - \mu) - (x_e - \mu))^2), \mu \triangleq E(x) \\ &= E((x - \mu)^2) + E((x_e - \mu)^2) \quad \Rightarrow \\ &\quad - 2E((x - \mu)(x_e - \mu)) \\ &= V(x) + (x_e - \mu)^2 \\ \hat{x} &\triangleq \arg \min_{x_e} L(x_e) = \mu \end{aligned}$$

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Estimation from a multi dimensional distribution

Given the density function $f(x)$ for a stochastic vector x find the estimate $\hat{x}$ that minimize the mean square error (MSE) $E((x - x_e)^T M (x - x_e))$ where x_e is any estimate and $M \geq 0$.

$$\begin{aligned} L(x_e) &= E((x - x_e)^T M (x - x_e)) \\ &= E(((x - \mu) - (x_e - \mu))^T M ((x - \mu) - (x_e - \mu))) \\ &= E((x - \mu)^T M (x - \mu)) + (x_e - \mu)^T M (x_e - \mu) \\ &= \text{trace}(M \Sigma) + (x_e - \mu)^T M (x_e - \mu) \\ &\quad , \Sigma \triangleq V(x) \triangleq E((x - \mu)^T (x - \mu)) \Rightarrow \end{aligned}$$

$$\hat{x} \triangleq \arg \min_{x_e} L(x_e) = \mu$$

Estimating one vector given another

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Estimation from a conditional distribution

Given the *conditional* density function $f(x|y)$ for a stochastic vector x given y find the estimate $\hat{x}$ that minimize the *conditional* mean square error (MSE) $E((x - x_e)^T M (x - x_e) | y)$ where x_e is any estimate based on y and $M \geq 0$.

- ▶ The *conditional* distribution is still just a distribution.
- ▶ The solution is therefore the same as for other distribution i.e. the mean value which now is a conditional mean

$$L(x_e) = E \left((x - x_e)^T M (x - x_e) | y \right) \Rightarrow$$

$$\hat{x} \triangleq \arg \min_{x_e} L(x_e) = E(x | y)$$

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Using unconditional loss

The conditional mean estimator will also minimize the *unconditional* loss

$$\begin{aligned} L_u(x_e) &\triangleq \mathbb{E} \left((x - x_e)^T M (x - x_e) \right) \\ &= \mathbb{E} \left(\mathbb{E} \left((x - x_e)^T M (x - x_e) | y \right) \right) \end{aligned}$$

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The estimator $\hat{x} = \mathbb{E}(x|y)$ that minimizes the conditional MSE for any given y will also minimize the mean (with respect to y) MSE.

Notice:

- Above the general rule $\mathbb{E}(g(x, y)) = \mathbb{E}(\mathbb{E}(g(x, y)|y))$ was used.

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The error is unbiased

$$\begin{aligned} E(x - \hat{x}) &= E(E((x - \hat{x})|y)) \\ &= E(E((x - E(x|y))|y)) \\ &= E(\underline{0}) \\ &= \underline{0} \end{aligned}$$

Notice:

- An estimator with bias would be a bad estimator.

The error is uncorrelated with the known information
This is also called the orthogonality principle.

$$\begin{aligned} & \mathbb{E}((x - \hat{x})(y - \mathbb{E}(y))^T) \\ &= \mathbb{E}(\mathbb{E}((x - \hat{x})(y - \mathbb{E}(y))^T | y)) \\ &= \mathbb{E}(\mathbb{E}((x - \mathbb{E}(x|y)) | y) (y - \mathbb{E}(y))^T) \\ &= \mathbb{E}(\underline{0}(y - \mathbb{E}(y))^T) = \underline{0} \end{aligned}$$

Notice:

- ▶ correlation between $(x - \hat{x})$ and y would mean that there were more information to extract.
- ▶ correlation between $(x - \hat{x})$ and x is not zero.
- ▶ but correlation between $(x - \hat{x})$ and $\hat{x}$ is zero because $\hat{x}$ is a function of y .



Kalman filter for the linear DD problem

Necessary probability theory

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Conditional Gaussian distributions

Assume x, y is jointly Gaussian then $x|y$ is Gaussian with mean and covariance

$$E(x|y) = E(x) + \text{Cov}(x, y) \text{Cov}(y)^{-1} (y - E(y)) ,$$

$$\text{Cov}(x|y) = \text{Cov}(x) - \text{Cov}(x, y) \text{Cov}(y)^{-1} \text{Cov}(y, x) ,$$

$$\text{Cov}(x|y) \triangleq E(\tilde{x}\tilde{x}^T|y) = E(\tilde{x}\tilde{x}^T) , \quad \tilde{x} \triangleq x - E(x|y)$$

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Independent measurements

Assume x, u, v is jointly Gaussian and that u, v is independent then

$$E(x|u, v) = E(x|u) + E(x|v) - E(x)$$

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Notation

The notation used is not always the same as in Grewal.

$$\hat{x}_{k|k} \triangleq \hat{x}_k^{(+)} \triangleq E(x_k | \mathcal{Y}_k) ,$$

$$\hat{x}_{k|k-1} \triangleq \hat{x}_k^{(-)} \triangleq E(x_k | \mathcal{Y}_{k-1}) ,$$

$$\tilde{x}_{k|k} \triangleq \tilde{x}_k^{(+)} \triangleq x_k - \hat{x}_{k|k} ,$$

$$\tilde{x}_{k|k-1} \triangleq \tilde{x}_k^{(-)} \triangleq x_k - \hat{x}_{k|k-1} ,$$

$$P_{k|k} \triangleq P_k^{(+)} \triangleq E(\tilde{x}_{k|k} \tilde{x}_{k|k}^T) ,$$

$$P_{k|k-1} \triangleq P_k^{(-)} \triangleq E(\tilde{x}_{k|k-1} \tilde{x}_{k|k-1}^T)$$

Main idea

The derivation builds on the following (which would be hard to figure out on your own):

- ▶ The LMS estimator is $\hat{x}_k = E(x_k | \mathcal{Y}_k)$
- ▶ The state space model is recursive in time then the KF should also be recursive in time.
- ▶ Given the current estimate $\hat{x}_{k|k}$ it is easy to find predictions $\hat{x}_{k+j|k}$, $j \geq 1$ including the error covariance.
- ▶ The KF is split into a *measurement update* using y_k to update from $\hat{x}_{k|k-1}$ to $\hat{x}_{k|k}$ and a *time update* to update from $\hat{x}_{k|k}$ to $\hat{x}_{k+1|k}$ using no new measurements.

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Time update

Given $\hat{x}_{k|k}$ and $P_{k|k}$ calculate $\hat{x}_{k+1|k}$ and $P_{k+1|k}$

$$x_{k+1} = \Phi_k x_k + \Gamma_k u_k + w_k \Rightarrow$$

$$\hat{x}_{k+1|k} = \Phi_k \hat{x}_{k|k} + \Gamma_k u_k \Rightarrow$$

$$\tilde{x}_{k+1|k} = \Phi_k \tilde{x}_{k|k} + w_k \Rightarrow$$

$$P_{k+1|k} = \Phi_k P_{k|k} \Phi_k^T + Q_k .$$

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Given $\hat{x}_{k|k-1}$ and $P_{k|k-1}$ calculate $\hat{y}_{k|k-1}$ and $P_{k|k-1}^y$

$$y_k = H_k x_k + D_k u_k + v_k \Rightarrow$$

$$\hat{y}_{k|k-1} = H_k \hat{x}_{k|k-1} + D_k u_k \Rightarrow$$

$$\tilde{y}_{k|k-1} = H_k \tilde{x}_{k|k-1} + v_k \Rightarrow$$

$$P_{k|k-1}^y = H_k P_{k|k-1} H_k^T + R_k .$$

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Measurements update - estimate

Given $\hat{x}_{k|k-1}$, $P_{k|k-1}$ and $y(k)$ calculate $\hat{x}_{k|k}$ and $P_{k|k}$

$$\begin{aligned}\hat{x}_{k|k} &= E(x_k | \mathcal{Y}_k) \\ &= E(x_k | y_k, \mathcal{Y}_{k-1}) \\ &= E(x_k | \tilde{y}_{k|k-1}, \mathcal{Y}_{k-1}) \\ &= \hat{x}_{k|k-1} + E(x_k | \tilde{y}_{k|k-1}) - E(x_k) .\end{aligned}$$

$$\begin{aligned}E(x_k | \tilde{y}_{k|k-1}) &= E(x_k) + \\ &\quad \text{Cov}(x_k, \tilde{y}_{k|k-1}) \text{Cov}(\tilde{y}_{k|k-1})^{-1} \tilde{y}_{k|k-1}\end{aligned}$$

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$$\text{Cov}(x_k, \tilde{y}_{k|k-1})$$

$$= \text{Cov}(\hat{x}_{k|k-1} + \tilde{x}_{k|k-1}, \tilde{y}_{k|k-1})$$

$$= \text{Cov}(\hat{x}_{k|k-1}, \tilde{y}_{k|k-1}) + \text{Cov}(\tilde{x}_{k|k-1}, \tilde{y}_{k|k-1})$$

$$= \text{Cov}(\tilde{x}_{k|k-1}, \tilde{y}_{k|k-1})$$

$$= \text{E}(\tilde{x}_{k|k-1} \tilde{y}_{k|k-1}^T)$$

$$= \text{E}(\tilde{x}_{k|k-1} (H_k \tilde{x}_{k|k-1} + v_k)^T)$$

$$= \text{E}(\tilde{x}_{k|k-1} \tilde{x}_{k|k-1}^T H_k^T)$$

$$= P_{k|k-1} H_k^T$$

Collecting the above

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} \tilde{y}_{k|k-1}$$

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Measurements update - error variance

$$K_k \triangleq P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} \Rightarrow$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} \Rightarrow$$

$$\tilde{x}_{k|k} = \tilde{x}_{k|k-1} - K_k \tilde{y}_{k|k-1}$$

$$= \tilde{x}_{k|k-1} - K_k (H_k \tilde{x}_{k|k-1} + v_k)$$

$$= (I - K_k H_k) \tilde{x}_{k|k-1} - K_k v_k \Rightarrow$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^T + K_k R_k K_k^T$$

$$= (I - K_k H_k) P_{k|k-1}$$

Kalman filter algorithm

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Initial conditions:

$$x_0 \in N(\hat{x}_{0|-1}, P_{0|-1})$$

Measurements update after receiving y_k and u_k :

$$\hat{y}_{k|k-1} = H_k \hat{x}_{k|k-1} + D_k u_k ,$$

$$\tilde{y}_{k|k-1} = y_k - \hat{y}_{k|k-1} ,$$

$$K_k \triangleq P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} ,$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^T + K_k R_k K_k^T .$$

Time update from k to $k + 1$:

$$\hat{x}_{k+1|k} = \Phi_k \hat{x}_{k|k} + \Gamma_k u_k ,$$

$$P_{k+1|k} = \Phi_k P_{k|k} \Phi_k^T + Q_k .$$

Multi step prediction based on Kalman filters

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Initial conditions:

Assume $\hat{x}_{k|k}, P_{k|k}$ know,

Multi step prediction is then obtained for $j \geq 0$ by:

$$\hat{x}_{k+j+1|k} = \Phi_{k+j} \hat{x}_{k+j|k} + \Gamma_{k+j} u_{k+j} ,$$

$$P_{k+j+1|k} = \Phi_{k+j} P_{k+j|k} \Phi_{k+j}^T + Q_{k+j} ,$$

$$\hat{y}_{k+j+1|k} = H_{k+j+1} \hat{x}_{k+j+1|k} + D_{k+j+1} u_{k+j+1} ,$$

$$P_{k+j+1|k}^y = H_{k+j+1} P_{k+j+1|k} H_{k+j+1}^T + R_{k+j+1} .$$

Multi step prediction can be interpreted as: simulation starting with initial state $\hat{x}_{k|k}$

Block diagram interpretation - mean value part (no direct feed though)

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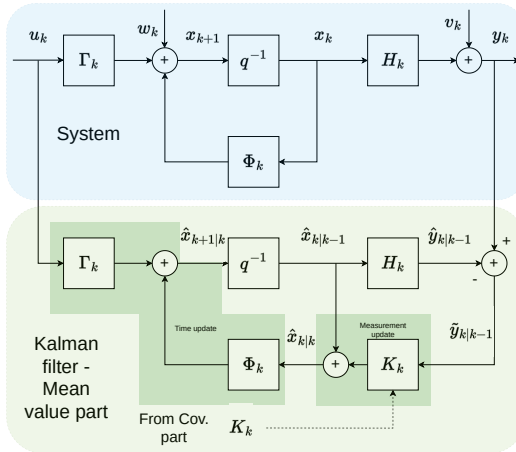
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Block diagram interpretation - covariance value part

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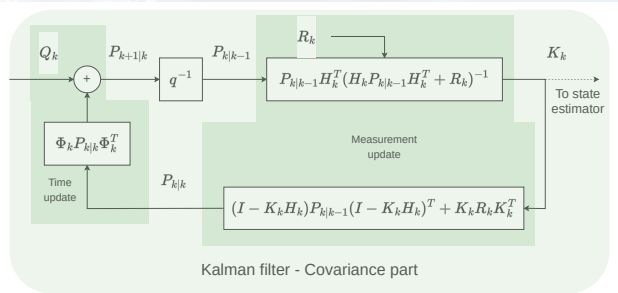
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Other important features of Kalman filter

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- ▶ The filter gain K_k and error covariances $P_{k|k}, P_{k|k-1}, P_{k|k-1}^y$ does not depend on measurements.
- ▶ If the model is time invariant the Kalman filter also converges to a time invariant filter given by the solution to the *discrete time Riccati equation*

$$P \triangleq \lim_{k \rightarrow \infty} P_{k|k-1} ,$$

$$P = \Phi P \Phi^T - \Phi P H^T (H P H^T + R)^{-1} H P \Phi^T + Q$$

- ▶ The measurement errors $\tilde{y}_{k|k-1}, \tilde{y}_{j|j-1}, k \neq j$ are independent and uncorrelated because for e.g. $j < k$, $\tilde{y}_{k|k-1}$ is independent of $\mathcal{Y}_{k-1}$ and $\tilde{y}_{j|j-1}$ is a linear function of $\mathcal{Y}_j \subseteq \mathcal{Y}_{k-1}$ for $j < k$.

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Possibilities with the Kalman filter

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The Kalman filter is a fantastic flexible tool

- ▶ The Kalman filter can serve many purposes.
- ▶ Creative use of the parameters makes it possible to tailor the Kalman filter to the problem.
- ▶ State estimation, smoothing and prediction is the basic application.
- ▶ Parameters estimation for ARX models is a less standard problem that can be solved.

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Challenges using the Kalman filter

Kalman filter

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State
estimation part
of the course

State
estimation
problem for
linear systems

Estimators
based on a
probabilistic
models

Kalman filter
for the linear
DD problem

Possibilities
with the
Kalman filter

Challenges
using the
Kalman filter

- ▶ If the process and measurement noise is not white the filter is not optimal.
 - ▶ Colored noise is included by augmenting the original state space model.
 - ▶ Correlation between process and measurement noise can also be included in the Kalman filter.
- ▶ All parameters must be known.
- ▶ Especially noise covariances are difficult to get.
 - ▶ Some times they can be found from *first principles*.
 - ▶ If input, output and states can be measured they can be estimated.
 - ▶ If only input and output, but not states, can be measured only part of the covariance parameters can be estimated.
 - ▶ Noise covariances can also be interpreted as tuning parameters.
- ▶ The system must of course be *observable* for state estimates to gain from measurements.